

RESEARCH AND EDUCATION

Strength and phase transformation of different zirconia types after chairside adjustment



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The computer-aided design and computer-aided manufacturing (CAD-CAM) technique has the ability to produce zirconia restorations with sufficient precision, strength, and appearance for dental use.¹ Zirconia can be used for single-tooth restorations, fixed dental prostheses, and implant-supported restorations.² Zirconia can exist in 3 phases depending on temperature: monoclinic (room temperature to 1170 °C), tetragonal phase (1170 °C to 2370 °C), and cubic (2370 °C to the 2690 °C melting point).³ The tetragonal form can be stabilized at room temperature when doped with yttrium oxide (Y₂O₃). This doping increases the mechanical properties, providing the main advantage of yttria-stabilized zirconia.⁴

Zirconia can be classified into 3 types based on the mol % of yttria: 3% yttria-partially stabilized zirconia (3Y-PSZ),

ABSTRACT

Statement of problem. Limited evidence is available for the effect of chairside adjustment using diamond instruments on different types of zirconia.

Purpose. The purpose of this in vitro study was to evaluate the effect of simulated adjustments on the biaxial flexural strength and phase transformation of 3 different zirconia types.

Material and methods. Three zirconia types from the same manufacturer (Katana; Kuraray) were used: High Translucency (3Y-PSZ), Super Translucent Multi Layered (4Y-PSZ), and Ultra Translucent Multi Layered (5Y-PSZ). Thirty disk-shaped specimens (Ø14×1.2 mm) were fabricated according to the International Organization for Standardization (ISO) Standard 6872 from different zirconia types (N=90). Specimens were either left without adjustment (NA), adjusted with Dialite ZR finishing and polishing system (Brasseler) (APol), or adjusted with course diamond instruments only (ADia). The specimens were distributed into 9 groups (n=10): group 3Y-PSZ/NA, group 3Y-PSZ/APol, group 3Y-PSZ/ADia, group 4Y-PSZ/NA, group 4Y-PSZ/APol, group 4Y-PSZ/ADia, group 5Y-PSZ/NA, group 5Y-PSZ/APol, and group 5Y-PSZ/ADia. The biaxial flexural strength of each specimen was measured by using a universal testing machine (Model 4411; Instron) and according to ISO 6872. X-ray diffraction analysis was conducted to quantify the monoclinic phase transformation. Scanning electron microscopy images were obtained to evaluate the fracture pattern. Statistical analysis was performed by using analysis of variance (ANOVA) and Tukey multiple comparison tests for pairwise comparisons ($\alpha=.05$).

Results. The mean biaxial flexural strengths ranked from the highest to the lowest were for 3Y-PSZ, 4Y-PSZ, and 5Y-PSZ under any test condition ($P=.007$). Chairside adjustment with a diamond instrument significantly decreased the flexural strength of all zirconia types ($P<.05$). No statistically significant difference was found between the effect of APol and ADia on the strength of zirconia 3Y-PSZ ($P=.603$), 4Y-PSZ ($P=.993$), and 5Y-PSZ ($P=.660$). Phase transformation did not occur in the 5Y-PSZ groups. ADia groups had significantly higher phase transformation values regardless of zirconia type ($P<.05$).

Conclusions. The biaxial flexural strength of zirconia decreased significantly after chairside adjustment with diamond instruments regardless of the yttria percentage. Adjustment with the Dialite ZR finishing and polishing system caused less tetragonal to monoclinic phase transformation than adjustment with a course-grit diamond instrument. (J Prosthet Dent 2024;132:455-63)

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Clinical Implications

Chairside adjustment of zirconia may lead to deleterious effects. Therefore, zirconia adjustment should be minimized to the greatest extent possible.

4% yttria-partially stabilized zirconia (4Y-PSZ), and 5% yttria-partially stabilized zirconia (5Y-PSZ).⁵ Of these, 3Y-PSZ is the strongest. However, it is opaque, which limits its dental use to nonesthetic zones.⁶ In contrast, 5Y-PSZ has better optical properties but inferior mechanical properties. 4Y-PSZ is considered an intermediate material providing a balance of acceptable optical and mechanical properties.⁷ However, translucent 5Y-PSZ zirconia has been considered to be a suitable option for the anterior region, meeting the minimum required strength and acceptable esthetics.⁶

Zirconia materials undergo a unique phase transformation from tetragonal-to-monoclinic phase when damage occurs, enhancing its strength and fracture toughness, a phenomenon called transformation toughening.⁸ Adjustment of indirect restorations before definitive cementation is a common clinical practice and may be required to obtain accurate proximal contacts, contours, and occlusion. However, adjustment may compromise zirconia's physical and mechanical properties.⁹⁻¹¹ Zirconia restorations of different types can be adjusted with different rotary cutting instruments at different rotational speeds of the instrument and for different adjustment amounts.¹² Clinicians may adjust zirconia restorations without clear guidance since evidence on how to properly adjust different types of zirconia intraorally is lacking.

The effect of adjustment on the mechanical properties of the 3 types of zirconia (3Y-PSZ, 4Y-PSZ, 5Y-PSZ) was evaluated by using 2 different chairside adjustments. The null hypotheses were that there would be no difference in flexural strength and phase transformation between different zirconia types before and after adjustment and that no difference would be found in the effect of different chairside adjustment protocols on the properties of different zirconia type.

MATERIAL AND METHODS

The materials used in this study are summarized in Table 1. According to the International Organization for Standardization (ISO) Standard 6872,¹³ 90 disk-shaped specimens (Ø14×1.2-mm) were milled (PM7 Milling Machine; Ivoclar AG) from disks of each zirconia type and sintered according to the manufacturer's instructions. After sintering, the 1.2 ±0.05 mm thickness of

the specimens was achieved by grinding with a sequence of 380-, 600-, and 1200-grit silicon carbide papers (MicroCut; Buehler) under running water. Final specimen dimensions were checked by precision digital calipers (Digimatic Micrometer; Mitutoyo Corp). Specimens were cleaned ultrasonically in distilled water for 5 minutes and air-dried for 20 seconds.

Three chairside adjustment protocols were evaluated: no adjustment (NA), adjustment complete steps (APol), and adjustment with course diamond only (ADia). The specimens were distributed to the test groups (n=10): group 3Y-PSZ/NA, group 3Y-PSZ/APol, group 3Y-PSZ/ADia, group 4Y-PSZ/NA, group 4Y-PSZ/APol, group 4Y-PSZ/ADia, group 5Y-PSZ/NA, group 5Y-PSZ/APol, and group 5Y-PSZ/ADia. The sample size was calculated by using a software program (G*Power v. 3.1.9.3; Heinrich-Heine-Universität Düsseldorf). Input values were set as effect size $f=0.40$, α error probability=0.05, and power (1- β error probability)=0.967. The total sample size was determined to be 90 (n=10/group). A zirconia finishing and polishing rotary instrument kit (Dialite ZR; Brasseler) was used for both APol and ADia. Adjustment of the specimens was done unidirectionally by a single operator (A.A.) with an electric handpiece (CA 1.5L; Bein-Air) with air-water coolant. For APol, the adjustment started with a fine-grit football-shaped diamond rotary instrument (Dialite 8369DF.31.025 FG; Brasseler USA) with a handpiece rotated at 30 000 rpm for 10 strokes followed by another 10 strokes using an extra-fine-grit football-shaped diamond rotary instrument (Dialite 369DEF.31.025 FG; Brasseler USA). Polishing was done using Intraoral Green Medium Cup (Dialite W17MZR.RA; Brasseler USA) followed by Orange Fine Cup (Dialite W17fZR.RA; Brasseler USA) at 6000 rpm until the specimen surface was visibly shiny. The polishing step took 60 to 80 seconds. For ADia, the electric handpiece was at 200 000 RPM with an air-water coolant with a Course Football Shaped Diamond (Dialite 6379.31.023 FG; Brasseler USA) for 10 strokes. No further adjustment or polishing was done.

All specimens were subjected to biaxial flexural strength testing with a universal testing machine (Model 4411; Instron) according to ISO standard 6872.¹³ The specimen rested on 3 symmetrical steel balls (Ø4.5 ±2 mm) positioned 120 degrees apart on a Ø11 ±1-mm support circle, and the load was applied to the center of the top surface by a Ø1.4 ±0.2-mm piston at a rate of 1 mm/min until fracture occurred. Fracture loads were recorded through the software program associated with the universal testing machine (TestWorks; MTS).

The biaxial flexural strength was calculated according to the following formula: $S = -0.2387P(X-Y)/d^2$, where S =biaxial flexural strength (MPa); P =fracture load (N); d =specimen disk thickness at fracture origin (mm); $X=(1+v)\ln(B/C)^2 + [(1-v)/2](B/C)^2$; $Y=(1+v)\ln(A/C)^2 + [(1-v)/2](A/C)^2$,

Table 1. Monolithic zirconia types tested

Product	Lot Number and Shrinkage Factor	Yttria Concentration	Manufacturer
Katana High Translucent Katana (HT)	DGDIG 1.254	3 mol%	Kuraray Noritake Inc
Katana Super Translucent Multi Layered Katana (STML)	EEXCC 1.225	4 mol%	Kuraray Noritake Inc
Katana Ultra Translucent Multi Layered (Katana UTML)	DSQRI 1.246	5 mol%	Kuraray Noritake Inc

where ν =Poisson coefficient (ceramic=0.25, ISO 6872); A=radius of support circle (mm); B=radius of loaded area (mm); and C=radius of specimen disk (mm).

The monoclinic phase was identified and quantified with an X-ray diffraction (XRD) analysis. An Empyrean Multipurpose X-ray Diffractometer equipped with Cu-K α (45 kV, 40 mA) radiation was used (Empyrean Nano Edition; Malvern Panalytical). Five specimens from each group were selected for the XRD analysis. The XRD spectra were collected from the top surface of the specimens over a 2θ range between 10 degrees and 90 degrees at a scan speed of 1 degree/min and a step size of 0.02 degrees. The data were analyzed by using an XRD analysis software program (HighScore Plus; Malvern Panalytical).

The volume fraction of the monoclinic phase (V_m) was calculated by using the formula by Toraya et al¹⁴: $V_m=1.311X_m/1+0.311X_m$; $X_m=[I_m(-111)+I_m(111)]/[I_m(-111)+I_m(111)]+I_t(101)$, where I_m and I_t are the intensity of the monoclinic and tetragonal peaks, respectively. The relative amount percentage of V_m was reported for every adjustment tested.

For scanning electron microscopy (SEM) (S-4700; Hitachi), the specimens were gold-sputtered (SCD 005; BAL-TEC) for 130 seconds at 15 mA to obtain an 80-Å layer. Micrographs were obtained, and an area of 100 μ m at the fractured surface of the specimen was evaluated at $\times 500$ magnification.

A repeated-measures analysis of variance was used to assess the TP, CR, and Lb% ($\alpha=.05$) by using a within-specimen factor (paired comparison before and after fatiguing of the same specimen) and between-specimen factors (according to zirconia type and thickness). Pair-wise interactions of material and thickness by intervention were initially included in the model.

A 2-way analysis of variance (ANOVA) was used to assess the biaxial fracture strength ($\alpha=.05$) by using between-specimen variables. The 2 independent variables were type of chairside adjustments and type of zirconia tested. The post hoc Tukey multiple comparison test was used for pairwise comparisons ($\alpha=.05$). A statistical software package (IBM SPSS Statistics for Windows, v26.0; IBM Corp) was used for statistical analysis.

Table 2. Mean and standard deviation of biaxial flexural strength (MPa) of different zirconia types after chairside adjustment

Zirconia Type	No Adjustment	Adjustment Pol	Adjustment Dia
Katana High Translucent (HT) (3Y)	985 $\pm$ 51 ^a	703 $\pm$ 38 ^b	667 $\pm$ 52 ^b
Katana Super Translucent Multi Layered (STML) (4Y)	602 $\pm$ 68 ^a	418 $\pm$ 18 ^b	422 $\pm$ 19 ^b
Katana Ultra Translucent Multi Layered (UTML) (5Y)	377 $\pm$ 34 ^a	156 $\pm$ 7 ^b	168 $\pm$ 7.61 ^b

Different superscripted letters in same row show significant differences ($P<.05$).

RESULTS

Mean and standard deviation values of the biaxial flexural strength (MPa) of all the tested groups are summarized in Table 2. The mean biaxial flexural strength ranged from the highest to the lowest in the following order at any test condition with statistical significance ($P=.007$): 3Y-PSZ, 4Y-PSZ, and 5Y-PSZ (Fig. 1). Adjustment had a significant effect on the mean biaxial flexural strength of all zirconia types ($P<.05$). No statistically significant difference was found between the APol and ADia effect on 3Y-PSZ ($P=.603$), 4Y-PSZ ($P=.993$), or 5Y-PSZ ($P=.660$).

XRD spectra of all tested groups are presented in Figure 2. The monoclinic phase peaks were observed only in the 3Y-PSZ and 4Y-PSZ groups. Relative percentages of V_m for each group are summarized in Table 3. When comparing the volume fraction of the monoclinic phase (V_m) between APol and ADia, the ADia groups had statistically significant higher values independent of zirconia type ($P<.05$).

Representative SEM micrographs are presented in Figure 3. In the NA groups, the fracture line was homogenous with minimum to no irregularities, resulting in single-plane fracture. In the APol and ADia groups, the fracture line was irregular with a scattered surface. The fracture lines originated from the scratches and surface irregularities caused by the adjustment procedure. The surface of the APol specimens was visibly smoother than that of the ADia specimens.

DISCUSSION

The effect of different chairside adjustments on the biaxial flexural strength and phase transformation of different zirconia types was tested in this study. The first null hypothesis was rejected since adjustment had a significant effect on the biaxial flexural strength and phase transformation of different types of zirconia. The second null hypothesis was accepted since no statistically significant difference exist between the effect of the 2 tested chairside adjustment protocols on zirconia properties.

Both adjusted groups had significantly lower biaxial flexural strength values than the unadjusted group,

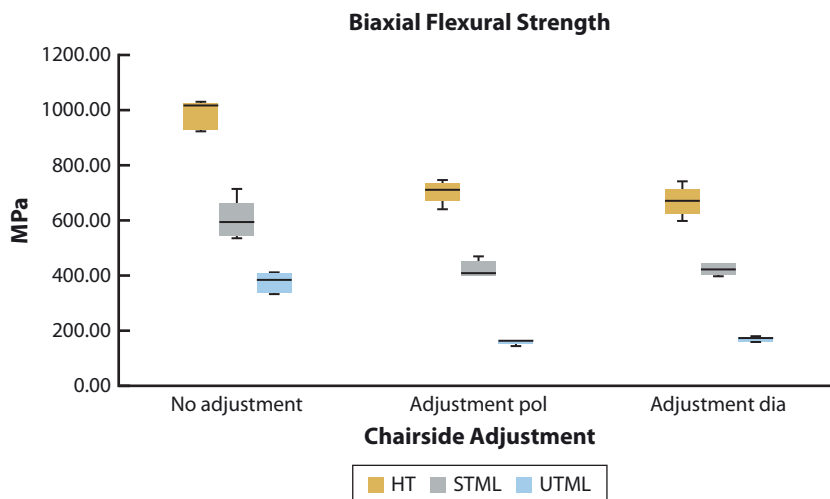


Figure 1. Comparison of biaxial flexural strength of different zirconia types after chairside adjustment. No adjustment, specimens untouched; adjustment Pol, specimens adjusted and polished following protocol; adjustment Dia, specimen adjusted with coarse diamond rotary instrument with no polishing. HT, High-Translucency Zirconia; STML, Super Translucent Multi Layered Zirconia; UTML, Ultra Translucent Multi Layered Zirconia.

independent of the zirconia type. The reduction ranged between 150 and 250 MPa, decreasing the biaxial flexural strength by 29% to 59%.

Michida et al¹⁵ reported a similar reduction of flexural strength of zirconia when treated with fine-grit diamond rotary instruments although they used the 4-point bend test. Kosmac et al¹⁶ also reported a reduction in zirconia strength when ground with diamond instruments only. However, Mohammadi-Bassir et al¹⁷ reported that grinding with diamond instruments increased the flexural strength of zirconia. However, they only tested 1 type of zirconia and performed 3-point flexural strength test. These studies did not test different types of zirconia and did not consider the yttria content.

The XRD analysis showed that 2 chairside adjustments tested led to tetragonal (t) → monoclinic (m) phase transformation for the 3Y and 4Y groups only, with no evidence of phase transformation in the 5Y groups. The lack of phase transformation could be explained by the high stabilization of 5Y zirconia because of the high yttria content (more cubic and less tetragonal particles). The ADia groups had more phase transformation than APol, possibly because rough adjustment left more cracks and surface defects, which might have initiated the tetragonal (t) → monoclinic (m) phase transformation. Polishing procedures undertaken in the APol group helped eliminate many surface cracks and produced a smoother surface. The SEM images confirmed more surface defects and irregularities for the ADia groups than for the APol groups. Karakoca and Yilmaz¹⁸ reported that grinding or airborne-particle abrading zirconia has no significant effect on phase transformation. Caglar et al¹⁹ tested multiple adjustment and polishing protocols on

zirconia specimens and concluded that none of them caused phase transformation. However, Mohammadi-Bassir et al¹⁷ indicated that grinding zirconia with coarse-grit diamond rotary instruments induced phase transformation, similar to the findings in the present study. Guilardi et al²⁰ concluded that grinding and aging increased the monoclinic phase content but did not affect the zirconia's mechanical properties. Unlike the present study, these 2 studies^{17,20} did not test different zirconia types.

The present study found that the introduction of surface defects and cracks during adjustment induced t → m phase transformation. When zirconia is adjusted, it is essential to consider the amount of yttria and its relationship with transformation toughening. In general, zirconia with a lower yttria content has more tetragonal particles capable of transforming to monoclinic and forming compressive strength around the crack tip, preventing it from propagating.²¹ As the amount of yttria content increases, the percentage of cubic particles increases, and that of tetragonal particles decreases.²¹ Cubic zirconia cannot undergo phase transformation, so crack propagation is easier, weakening the zirconia.²²

Although the effect of different adjustment methods on zirconia strength has been investigated, a standardized adjustment protocol has not been established. Factors such as the adjustment system type, application time, application load, and operator factors are challenging to control and could contribute to the disparities in results, making a firm conclusion difficult. In clinical practice, minimizing the chairside adjustment of zirconia restorations by optimizing laboratory procedures such as designing, milling, and polishing is recommended.

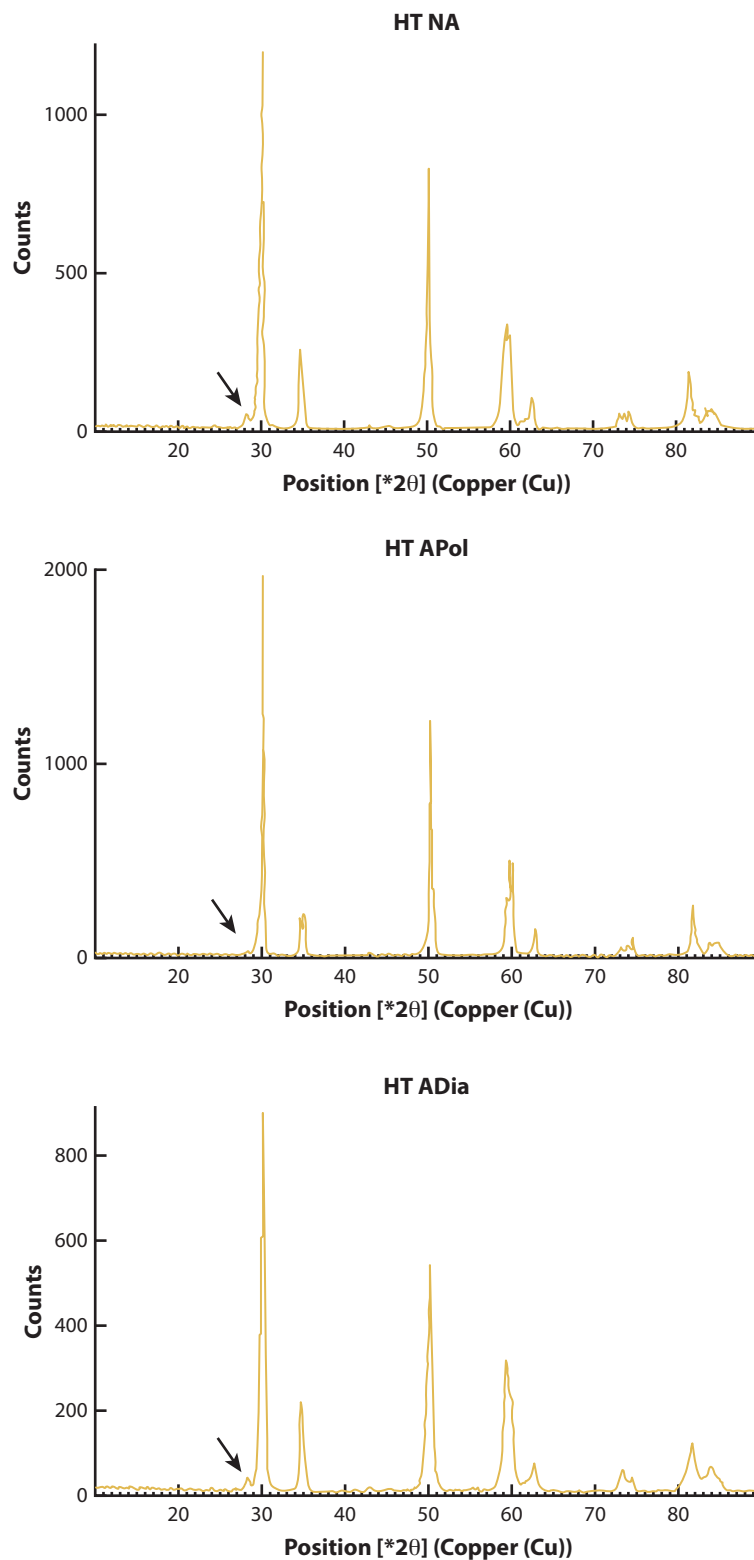


Figure 2. XRD spectra of all tested groups. Arrows indicating monoclinic peaks. ADia, adjustment Dia; APol, adjustment Pol; HT, High Translucency Zirconia; NA, no adjustment; STML, Super Translucent Multi Layered Zirconia; UTML, Ultra Translucent Multi Layered Zirconia; XRD, X-ray diffraction.

Limitations of the current study included that the zirconia tested came from a single manufacturer and that their behavior may differ from zirconia from other

manufacturers. The main goal, however, was to compare the different compositions of these materials, with a focus on yttria content. Another limitation was that flat

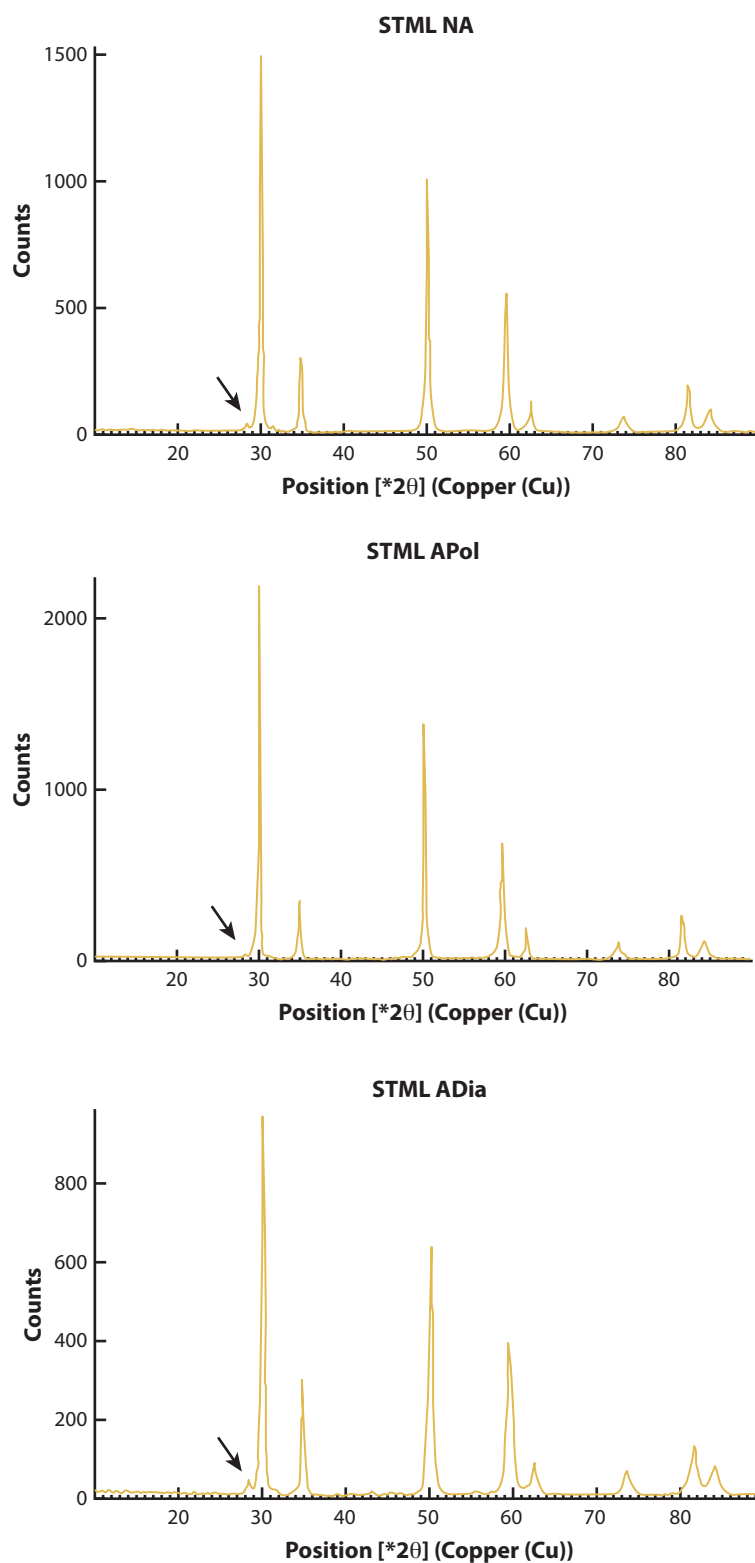


Figure 2. (continued)

unbonded specimens were tested. The focus was to test the inherent properties of the different types of zirconia and evaluate their crystallographic state. Clinically bonded

crowns might behave differently. The current study investigated the immediate effects, without thermo-mechanical aging, of chairside adjustment on the

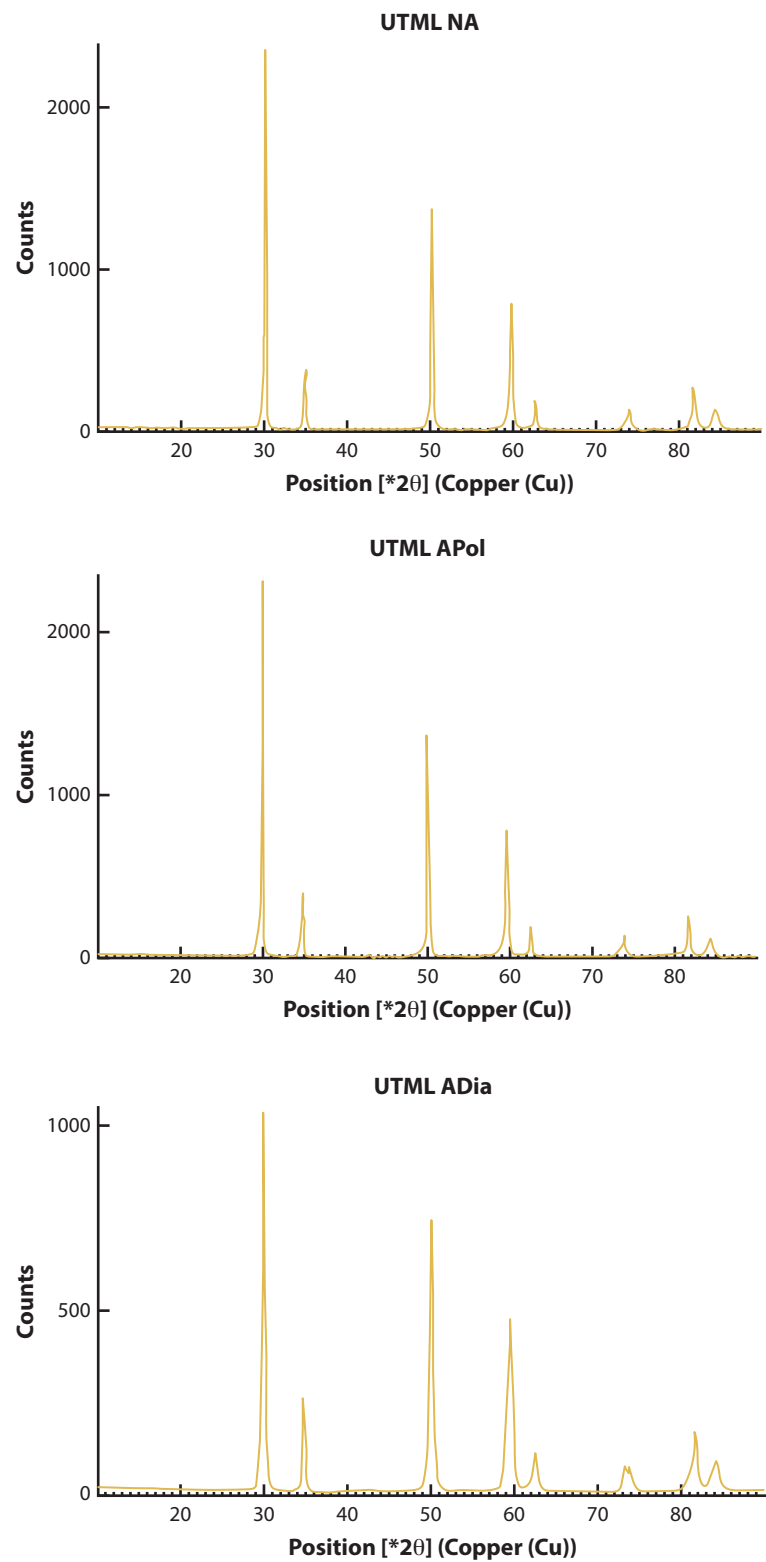


Figure 2. (continued)

properties of zirconia. Clinically relevant in vitro and in vivo studies are recommended to further investigate the effect of chairside adjustment on the properties of zirconia.

CONCLUSIONS

Based on the findings of this in vitro study, the following conclusions were drawn:

Table 3. Relative amount of Vm

Zirconia Type	Adjustment Pol	Adjustment Dia
Katana High Translucent (HT) (3Y)	1.95 ^a	3.6 ^b
Katana Super Translucent Multi Layered (STML) (4Y)	1.7 ^a	3.9 ^b
Katana Ultra Translucent Multi Layered (UTML) (5Y)	-	-

Different superscripted letters in same row show significant differences ($P<.05$).

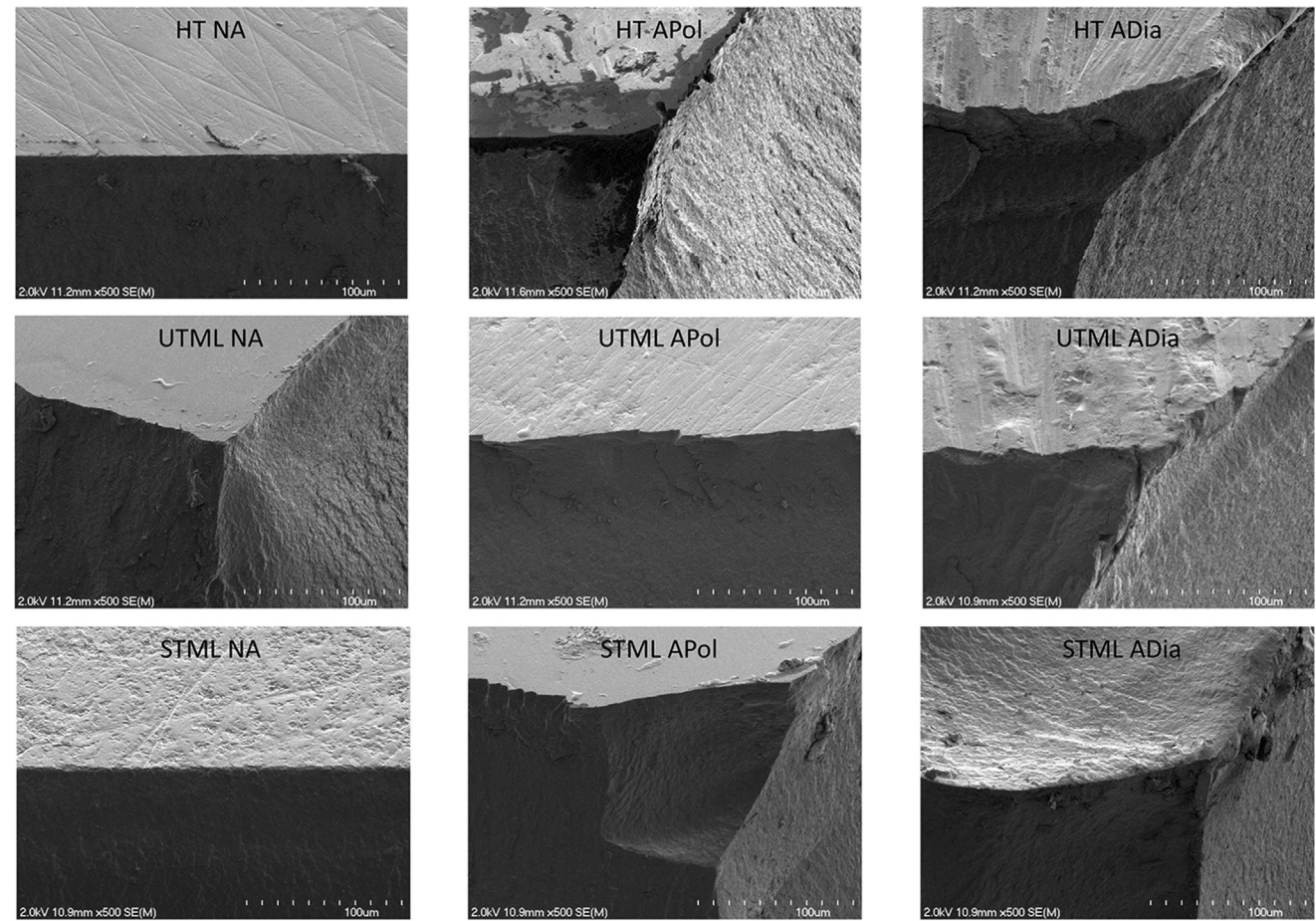


Figure 3. Representative SEM micrographs (original magnification $\times 500$) of all groups. ADia, adjustment Dia; APol, adjustment Pol; HT, High Translucency Zirconia; NA, no adjustment; STML, Super Translucent Multi Layered Zirconia; UTML, Ultra Translucent Multi Layered Zirconia.

1. Chairside adjustment with diamond instruments significantly reduced the biaxial flexural strength of zirconia and caused phase transformation depending on the yttria content of the zirconia.
2. Given any test condition, as the yttria content increased, the strength of the zirconia decreased.
3. Appropriate adjustment and finishing protocol for zirconias will cause less phase transformation than adjustments with a coarse diamond instrument alone.

REFERENCES

1. Boitelle P, Mawussi B, Tapie L, Fromentin O. A systematic review of CAD/CAM fit restoration evaluations. *J Oral Rehabil.* 2014;41:853–874.
2. Nistor L, Grădinaru M, Rică R, Mărășescu P, Stan M, Manolea H, et al. Zirconia use in dentistry - manufacturing and properties. *Curr Health Sci J.* 2019;45:28–35.
3. Denry I, Kelly JR. State of the art of zirconia for dental applications. *Dent Mater.* 2008;24:299–307.
4. Porter DL, Heuer AH. Mechanisms of toughening partially stabilized zirconia (PSZ). *J Am Ceram Soc.* 1977;60:183–184.
5. Burgess JO. Zirconia: the material, its evolution, and composition. *Compend Contin Educ Dent.* 2018;39:4–8.
6. Sulaiman TA, Abdulmajeed AA, Donovan TE, Ritter AV, Vallittu PK, Närhi TO, et al. Optical properties and light irradiance of monolithic zirconia at variable thicknesses. *Dent Mater.* 2015;31:1180–1187.
7. Ban S. Chemical durability of high translucent dental zirconia. *Dent Mater.* 2020;39:12–23.
8. Garvie RC, Hannink RH, Pascoe RT. Ceramic steel? *Nature.* 1975;258:703.
9. Lee KR, Choe HC, Heo YR, Lee JJ, Son MK. Effect of different grinding burs on the physical properties of zirconia. *J Adv Prosthodont.* 2016;8:137–143.
10. Guess PC, Zhang Y, Kim JW, Rekow ED, Thompson VP. Damage and reliability of Y-TZP after cementation surface treatment. *J Dent Res.* 2010;89:592–596.
11. Subaşı MG, Demir N, Kara Ö, Ozturk AN, Öz F. Mechanical properties of zirconia after different surface treatments and repeated firings. *J Adv Prosthodont.* 2014;6:462–467.
12. Sandhu R, Kheur M, Kheur S. Effect of simulated chairside grinding procedures using commercially available abrasive agents

- on the surface properties of zirconia. *J Indian Prosthodont Soc.* 2017;17:22–28.
13. International Organization for Standardization. ISO 9693-1. Dentistry compatibility testing. Part 1: metal-ceramic systems. Geneva: International Organization for Standardization; 2012. ISO Store Order: OP-184149 (Date: 2017-06-09). Available at: <http://www.iso.org/iso/home.html>.
 14. Toraya H, Yoshimura M, Sōmiya S. Quantitative analysis of monoclinic-stabilized cubic ZrO₂ systems by X-ray diffraction. *J Am Ceram Soc.* 1984;67: 183–184.
 15. Michida SM, Kimpara ET, dos Santos C, Souza RO, Bottino MA, Özcan M. Effect of air-abrasion regimens and fine diamond bur grinding on flexural strength, Weibull modulus and phase transformation of zirconium dioxide. *J Appl Biomater Funct Mater.* 2015;13:e266–e273.
 16. Kosmac T, Oblak C, Jevnikar P, Funduk N, Marion L. The effect of surface grinding and sandblasting on flexural strength and reliability of Y-TZP zirconia ceramic. *Dent Mater.* 1999;15:426–433.
 17. Mohammadi-Bassir M, Babasafari M, Rezvani MB, Jamshidian M. Effect of coarse grinding, overglazing, and 2 polishing systems on the flexural strength, surface roughness, and phase transformation of yttrium-stabilized tetragonal zirconia. *J Prosthet Dent.* 2017;118:658–665.
 18. Karakoca S, Yilmaz H. Influence of surface treatments on surface roughness, phase transformation, and biaxial flexural strength of Y-TZP ceramics. *J Biomed Mater Res B Appl Biomater.* 2009;91:930–937.
 19. Caglar I, Ates SM, Yesil Duymus Z. The effect of various polishing systems on surface roughness and phase transformation of monolithic zirconia. *J Adv Prosthodont.* 2018;10:132–137.
 20. Guilardi LF, Pereira GKR, Gündel A, Rippe MP, Valandro LF. Surface micro-morphology, phase transformation, and mechanical reliability of ground and aged monolithic zirconia ceramic. *J Mech Behav Biomed Mater.* 2017;65:849–856.
 21. Zhang Y, Lawn BR. Novel zirconia materials in dentistry. *J Dent Res.* 2018;97: 140–147.
 22. Zhang F, Inokoshi M, Batuk M, Hadermann J, Naert I, VanMeerbeek B, et al. Strength, toughness and aging stability of highly-translucent Y-TZP ceramics for dental restorations. *Dent Mater.* 2016;32:e327–e337.

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<https://doi.org/10.1016/j.prosdent.2022.06.015>